Waste Management in Molasses Based Distilleries

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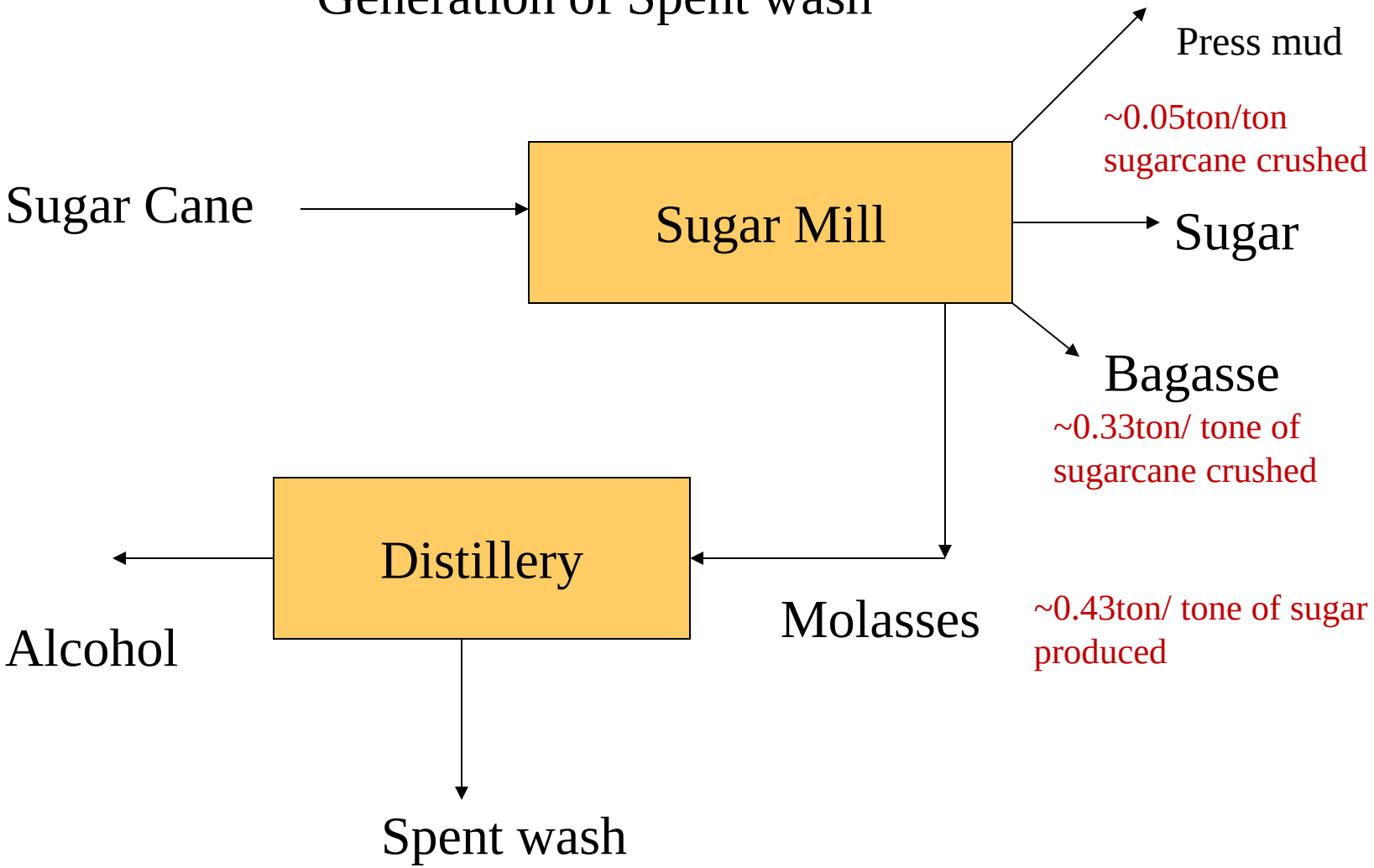
Industrial wastewaters Vs Sewage

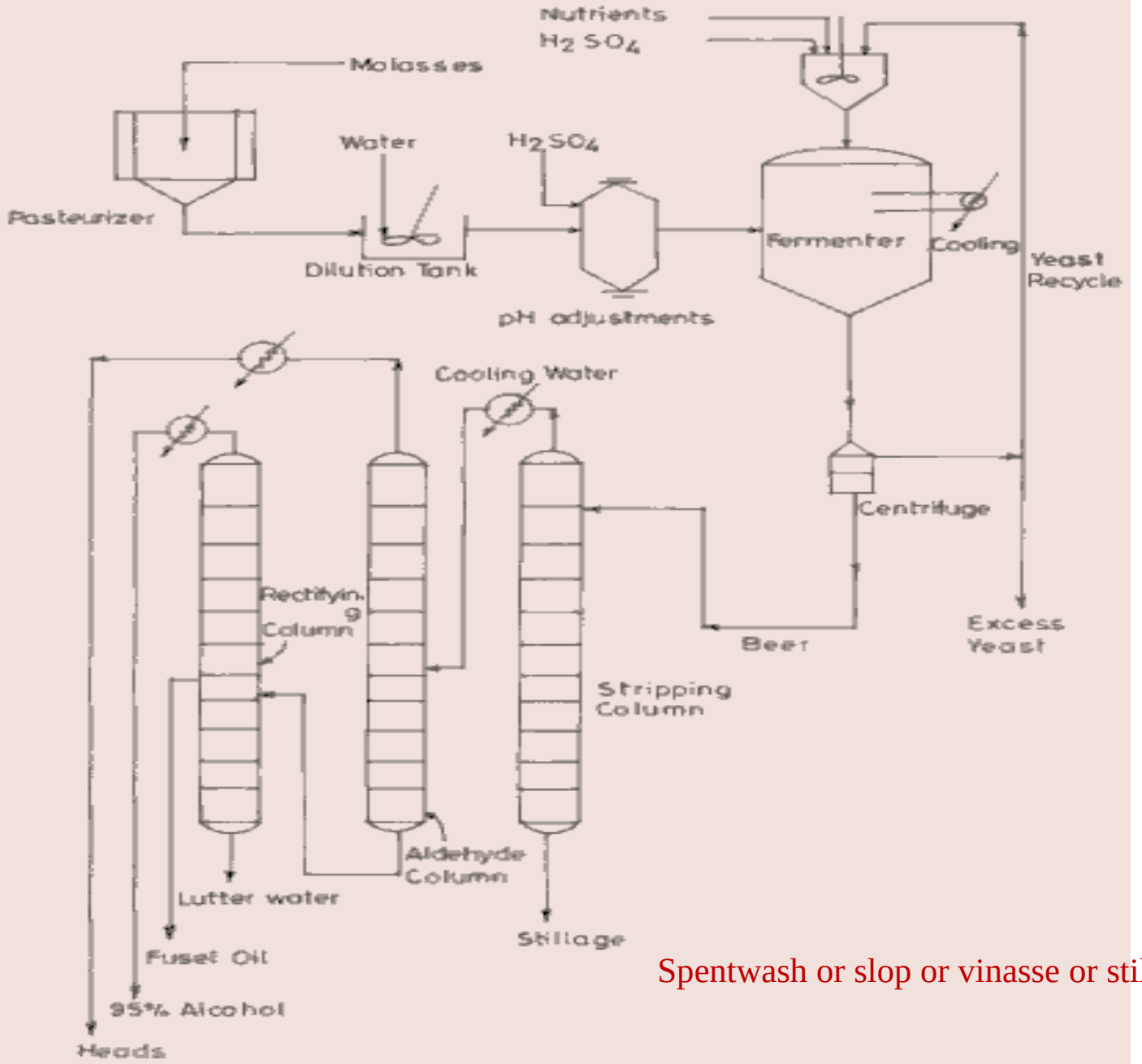
Generation of Spent wash



- Press mud
  - Mixed with spent wash for composting
- Bagasse:
  - used in pulp and paper industries
  - also as fuel in sugar mills / distilleries
- Molasses:
  - For manufacture of industrial alcohol

Distillery Industry in India

- Number of sugar mills: 620
- Molasses based distilleries: 300
- Molasses based installed capacity : 4.29 bln lit.
- Grain based distillery units: 100
- Grain based installed capacity : 1.8 bln lit.
- Molasses based distilleries produce spent wash having very high pollution load



Flow Sheet For a Molasses Fermentation and Distillation Unit (Sheehan and Greenfield, 1980)

Spentwash or slop or vinasse or stillage

Yeast propagation: Aerobic. Growth of yeast cells.

Fermentation: Sugar (10-20%) is converted to alcohol & CO_2 anaerobically. Takes bout 15-20 Hours at 30-36 ^oC (cooling required to maintain this temp.)

Stripping Column: to separate components from liquid mixture.

Aldehyde column: to remove aldehydes (they are carcinogenic)

Rectification column: to remove fusel oil and lutter water from alcohol.

Molasses Distilleries

- Spent wash is the main waste stream
  - It has a BOD_5 of about 35,000 to 65,000mg/lit
  - COD of about 60,000 – 1,20,000 mg/lit
  - pH – acidic (4 – 5)
  - Colour- dark brown
  - Sulfates > 7000 mg/L
  - About 12-15% solids content
  - Ash contains Potash as K_2O

Target Pollutants

- Organic matter
- pH
- Colour

- Spent wash generation: about 14 to 16 litres per liter for alcohol produced
    - BOD about 50 000 mg/lit
    - COD about 95 000 mg/lit
- Pollution load of total spent wash generated is equivalent to about 6 times the total BOD generated by the entire population of India

Attempts with Spent Wash

- For washing sugar cane
- For diluting molasses
- For irrigating sugar cane fields
- For manufacturing cattle feed
- For manufacturing yeast / dry ice / etc

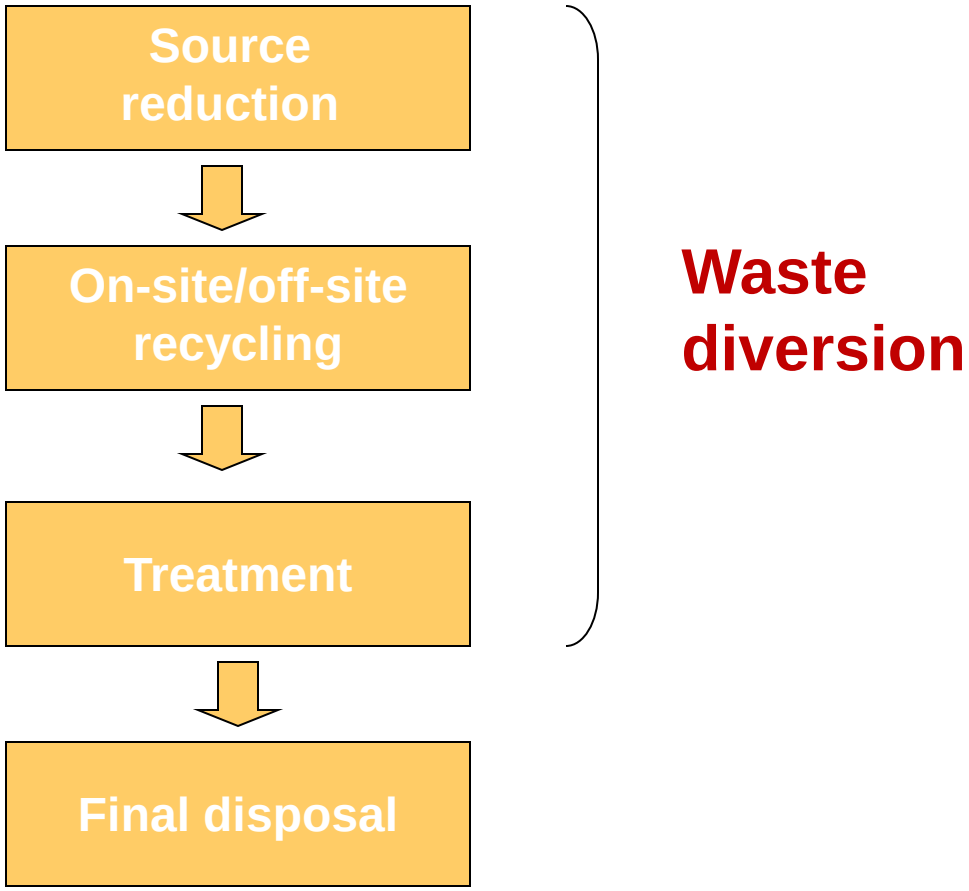
Nothing turned to be sustainable / feasible

Waste to be managed….

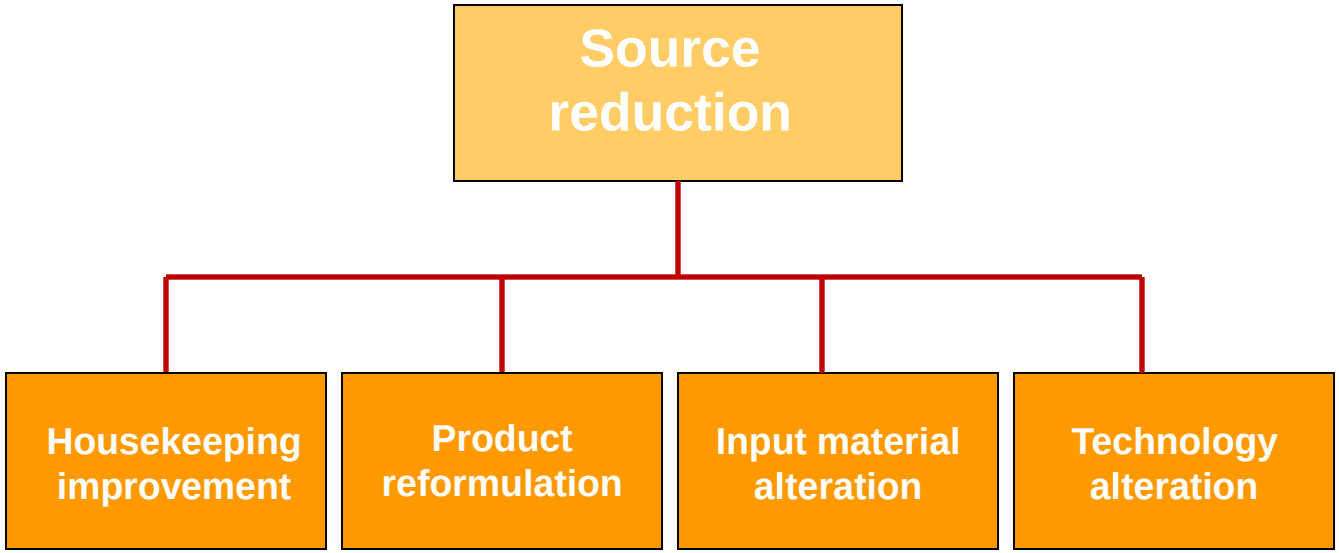
- Reduction at the source – especially by updating the fermentation unit

- Treatment - Disposal

Hierarchy of waste management options



Source reduction opportunities



Type of Fermentation units

- Batch and Fed-Batch
- Cascade Continuous
- Biostil Continuous

Cascade Continuous fermentation system

- Not one unit, but many
- Total 05 Fermentation units
  - 1st unit – mixing of culture, molasses and air, water
  - 2nd unit : aerobic – for yeast growth
  - 3rd to 5th units: anaerobic for sugar conversion to alcohol

  Alcohol % increases from 5% to 9% from fermenter 1 to 5

Biostil Continuous System

- Single fermenter with a recycle of about 60% of wash passed through the column for alcohol recovery for diluting molasses being fed to the fermenter
- Less water in the fermenter results in concentrated substrate
- Generates about 6-8 lit of spent wash /lit of alcohol produced. (15-17 lit /lit for conventional systems)
- Alcohol yield about 94.68% v/v

Average fresh water requirement in molasses based distilleries for different fermentation technologies.

Type of Fermentation Process (a)	Molasses Dilution fresh water (b)	Fresh Non process (RS Production)		Polluted Streams for recycle		Unpolluted Streams for recycle (g)	Total Streams for recycle (h)	Total fresh water require--ment (b+c+d-h)
		Cooling tower make up (c)	Other non process * (d)	For process (e)	For non process (f)			
Batch	**7.4 – 10.5**	**1.4 – 4.0**	**6.0 – 8.8**	**0.5 – 0.6**	**0.5 – 0.6**	**0.5 – 1.0**	**1.5 – 2.2**	**13.3–21.1**
Cascade Continuous	**7.7 – 9.2**	**3.0 – 4.0**	**5.5 – 6.5**	**0.5 – 1.5**	**0.5 – 0.6**	**0.5 – 1.0**	**1.5 – 3.0**	**14.7–16.7**
Biostill Continuous	**4.0 – 5.0**	**3.0 – 4.0**	**5.0 – 6.0**	**2.0 – 4.0**	**0.5 – 0.6**	**0.5 – 1.0**	**3.0 – 5.6**	**9.0 – 9.4**

* Other than cooling tower

Average waste water generation in molasses based distilleries for different fermentation technologies.

Type of Fermentation Process	Waste water generation		
	Spent wash	Spent lees	Other
Batch	11.1- 15.0	0.5 – 1.5	0.2 –0.4
Cascade Continuous	8.5 – 11.0	0.5 – 1.5	0.2 –0.4
Biostill Continuous	6.0 – 8.0	0.5 – 1.5	0.2 –0.4

General characteristics of raw spent wash

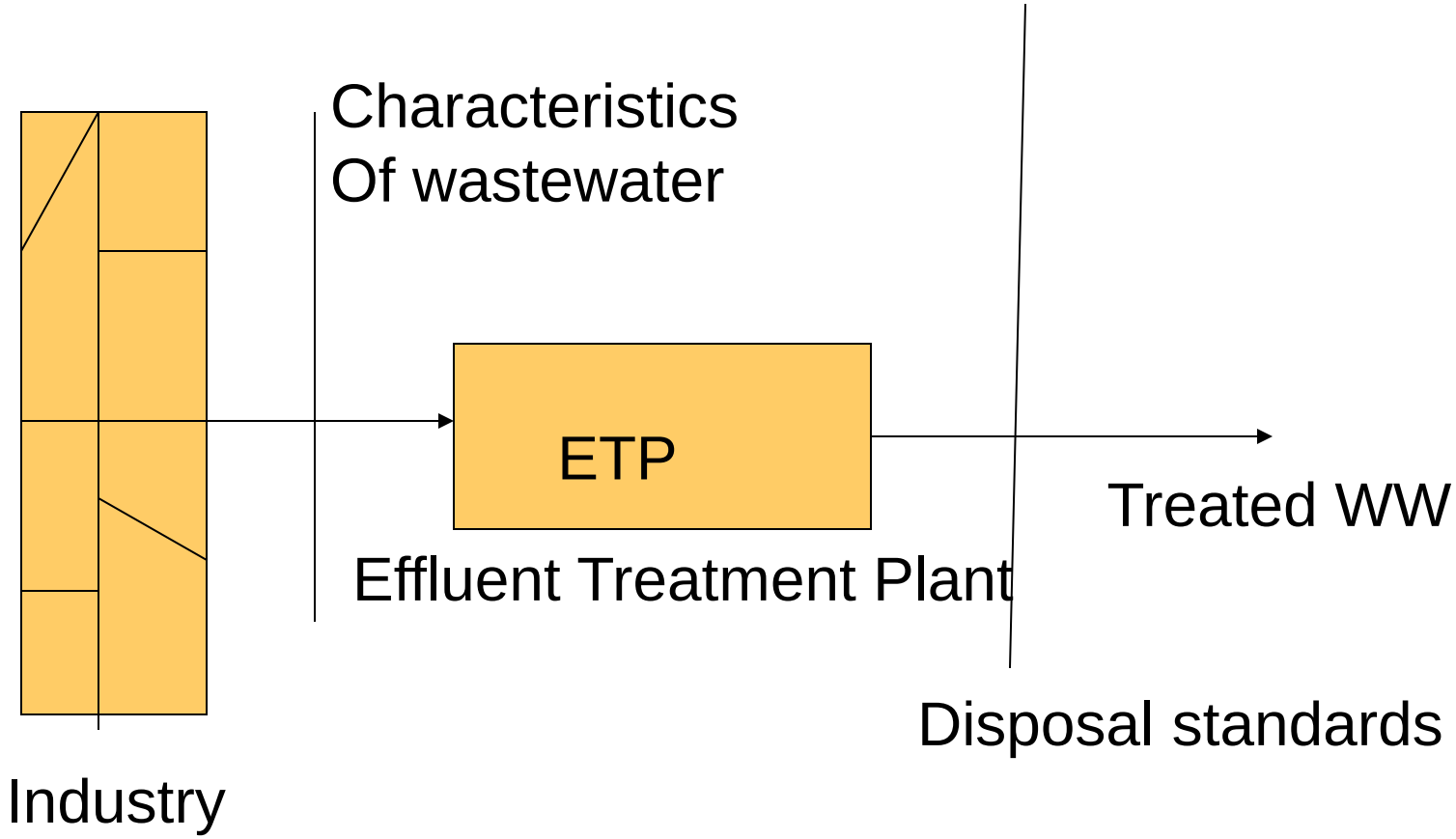
S.No.	Parameter	Batch process	Cascade process	Biostil process
1	Volume, L/L Alcohol	14-16	10-12	8-10
2	Colour	Dark brown	Dark brown	Dark brown
3	PH	3.7-4.5	4.0-4.3	4.0-4.2
4	COD	80,000-1,10,000	1,10,000-1,30,000	1,40,000-1,60,000
5	BOD	45,000-50,000	55,000-65,000	60,000-70,000
6	Solid Total Total Volatile Inorganic dissolved	90,000-1,20,000 60,000-70,000 30,000-40,000	1,30,000-1,60,000 60,000-75,000 35,000-45,000	1,60,000-2,10,000 80,000-90,000 60,000-90,000
7	Chlorides	5,000-6,000	6,000-7,500	10,000-12,000
8	Sulphates	4,000-8,000	4,500-8,500	8,000-10,000
9	Total nitrogen	1,000-1,200	1,000-1,400	2,000-2,500
10	Potassium	8,000-12,000	10,000-14,000	20,000-22,000
11	Phosphorus	200-300	300-500	1,600-2,000
12	Sodium	400-600	1,400-1,500	1,200-1,500
13	Calcium	2,000-3,500	4,500-6,000	5,000-6,500

- Spent wash generation: about 8 to 16 litres per litre for alcohol produced
    - 8-9 for new plants (Biostil)
    - 15-16 for old plants (batch)

Fermentation Industry : Effluent Discharge Standards (Distilleries, Maltries & Breweries) As notified under the Environment (Protection) Rule, 1986

Sr. No.	Parameter	Range
1	pH	5.5- 9.0
2	Colour & Odour	Absent
3	BOD3 (at 27°C, 3 days) mgs/L Max. Disposal into land surface water/ rivers/ streams Disposal on land or for irrigation.	 30 100
4	Suspended solids, mg/L Max.	100

Industrial Effluent Treatment



IS – for Effluent Discharge

PARAMETER	Surface water	Sewer lines	irrigation	ocean
Suspended solid	100mg/l	600mg/l	200mg/l	-
Dissolved solids	2100mg/l	-	2100mg/l	-
pH	5.5-9	5.5-9	5.5-9	5.5-9
Temperature	not>5^{0}Cthan the receiving water	-	-	Not>5 ^{0}C than the receiving water
Oil & greese	10mg/l	20mg/l	10mg/l	20mg/l
Free NH_3	5.0	-	-	5.0
$BOD_{5d,20^0C}$	30	350	100	100
COD	250	-	-	250
Arsenic	0.20	0.20	0.20	0.20
Hg	0.01	0.01	-	0.01
Pb	0.01	1.0	-	2.0
Cd	2.0	1.0	-	2.0
Cr^{+6}	0.1	2.0	-	1.0
Total Cr	2.0	2.0	-	2.0
Cu	3.0	3.0	-	3.0
Zn	5.0	15	-	15
Ni	3.0	3.0	-	5.0
Cyanide	0.2	2.0	0.2	0.2
Chlorides	1000	1000	600	-
Fluorides	2.0	15	-	15
Phosphates	5.0	-	-	-
Sulphates	1000	1000	1000	-
Sulphides	2.0	-	-	5.0
Phenolic compounds	1.0	5.0	-	5.0

Spent wash treatment

BOD/ COD is about 0.50-0.60

Hence biological treatment

It is high strength wastewater – hence anaerobic treatment

WASTE TREATMENT ROUTES

- Biological – for sure, THE BEST option
  - More eco-friendly
  - End products acceptable to the nature
  - Low expenditures
  - But slow, more uncertainty, affected by weather / temp

- Chemical – for lesser quantities

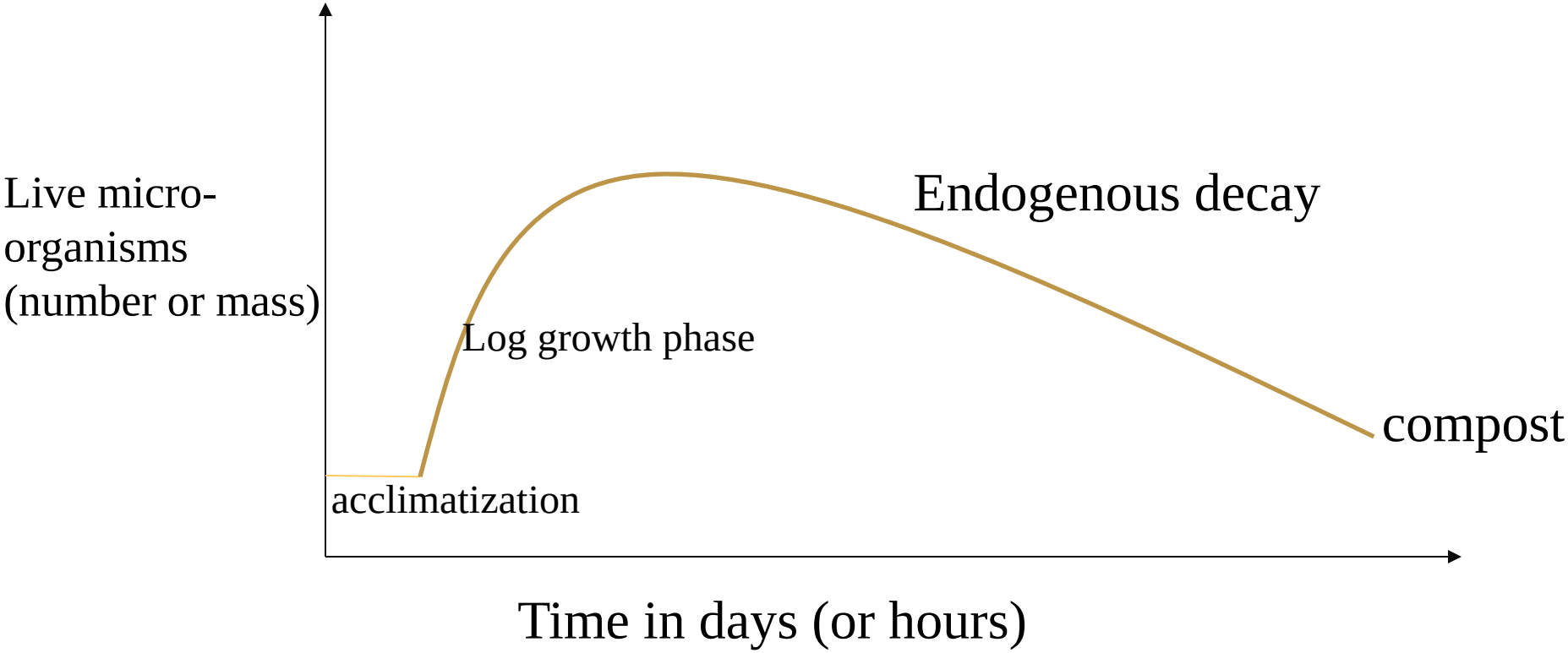
- Thermal-

  usually costlier, probability of pollution, controversial but very fast, compact reactors, less area required, more fool proof, not affected by weather / temperature, less uncertainty

Biological treatment

- Removal of organic matter using microorganisms (mostly bacteria, fungi, actinomycetis, etc)
- Organics used for energy and growth
- Organics get converted to gases (that escaped the system) and cell mass (has to be removed)

Batch reactor – growth of micro-organisams



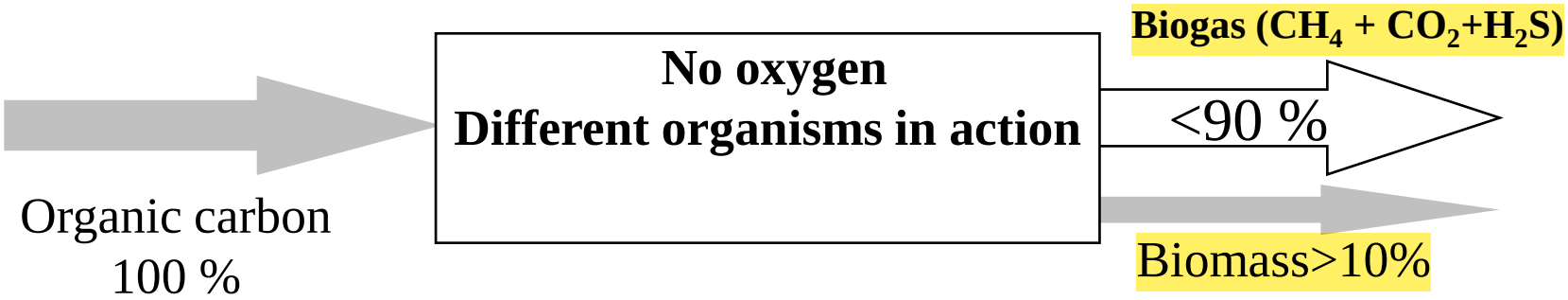
Biological treatment

- Aerobic: in the presence of Oxygen

$$\text{Organics} + O_2 \rightarrow CO_2 + H_2O + \text{More Biomass}$$

Biomass

- Anaerobic: in the absence of Oxygen



Biological reactors

- Batch or continuous:
  - Composting – batch process
  - Activated sludge process (ASP) - continuous
- Suspended growth or attached growth
  - Suspended growth: microorganisms are suspended in the waste water: ASP
  - Attached growth: microorganisms are attached to some media like pebbles, plastic balls, etc: Tricking filters, Rotating Biological Contactor (RBC)

BIOLOGICAL WASTEWATER TREATMENT

Aerobic:

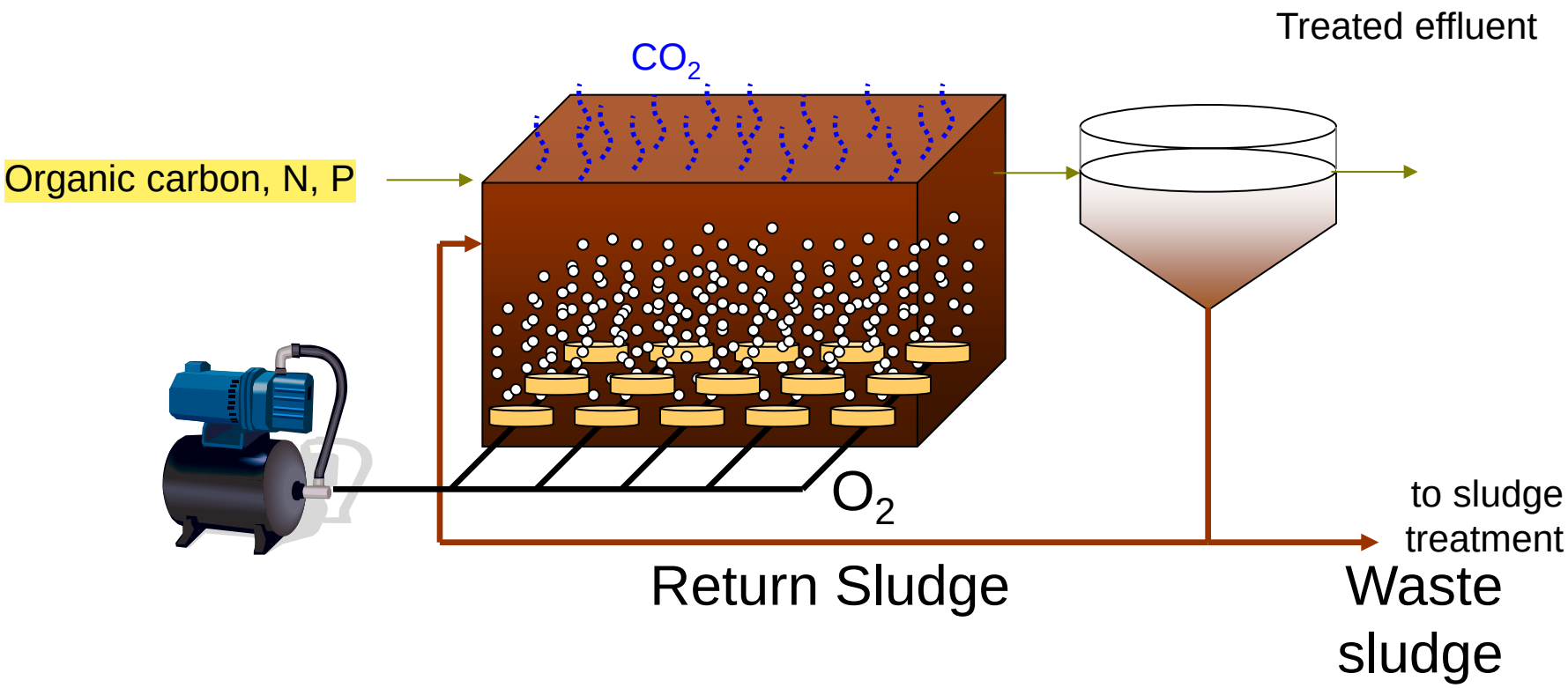
- ASP (Activated Sludge Process),
- TF (Trickling Filter),
- RBC (Rotating Biological Contactor)
- SBR (Sequential Batch Reactor)
- MBBR
- MBR

Anaerobic:

- Conventional Digester
- Di-phasic digestion
- UASB Reactor (Upflow Anaerobic Sludge Blanket Reactor)
- FB,
- Hybrid reactor

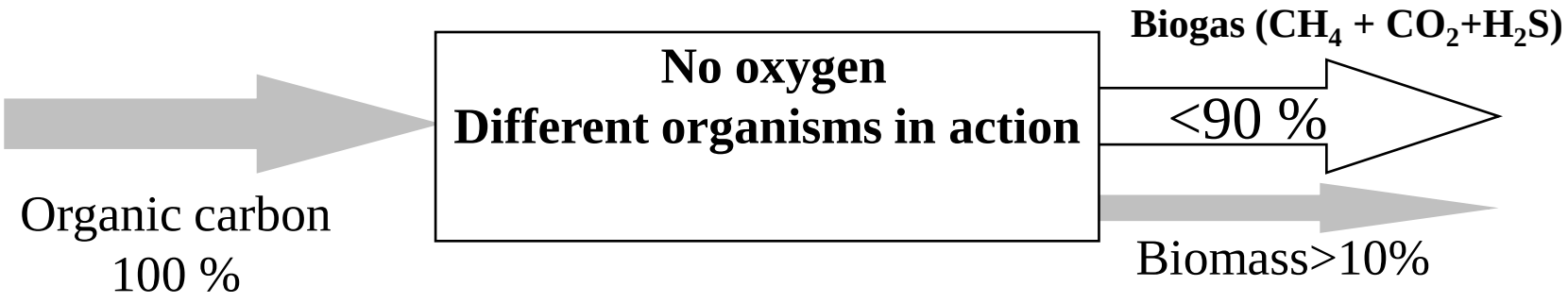
$$\text{Organics} + O_2 \longrightarrow CO_2 + H_2O + \text{More Biomass}$$

Biomass



ACTIVATED SLUDGE PROCESS (ASP)

Anaerobic process



Comparing aerobic – anaerobic techniques

- Aerobic
  - Faster reaction kinetics
  - Hence smaller reactors
  - No bad odour

  But
  - have to provide Oxygen
  - No any recovery,
  - more sludge to be handled

- Anaerobic

  - Fuel Gas recovery
  - Less sludge to be handled
  - No oxygn to be supplied

  But

  - Slow reaction kinetics
  - Large reactors
  - Odour issues are there

The main treatment strategy

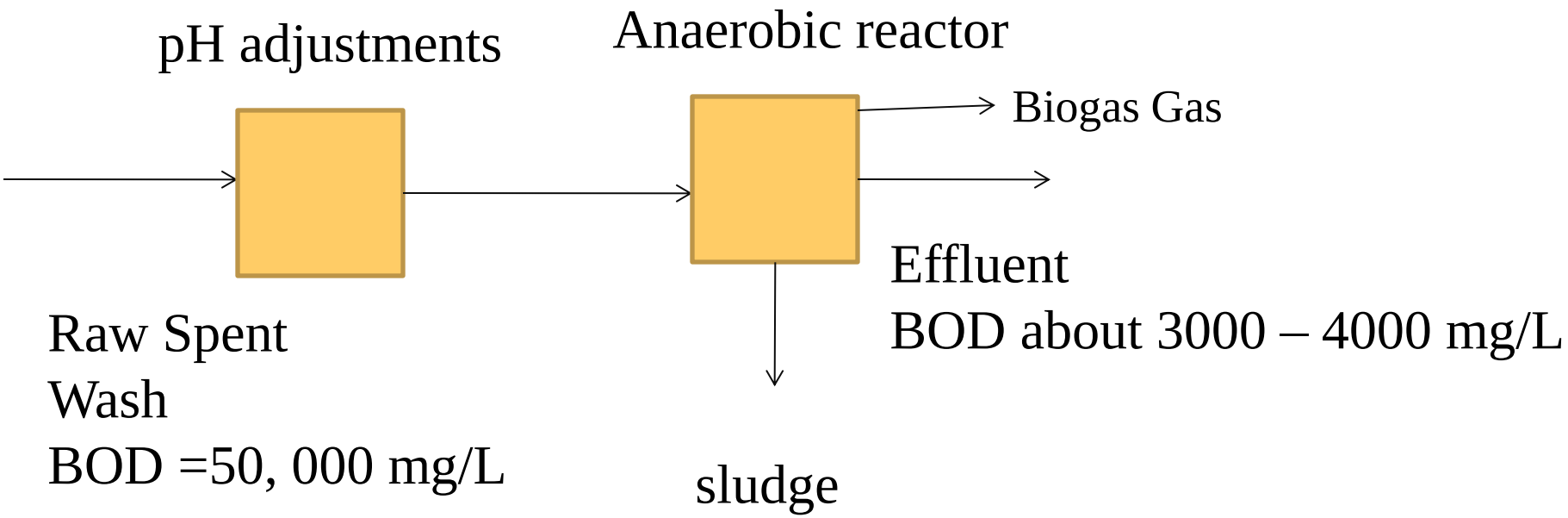
- BOD/COD = 55,000/1,00,000 = > 0.5

Hence biological treatment is effective

Since it is **high strength waste water**, anaerobic treatment technique is better

Spent wash treatment

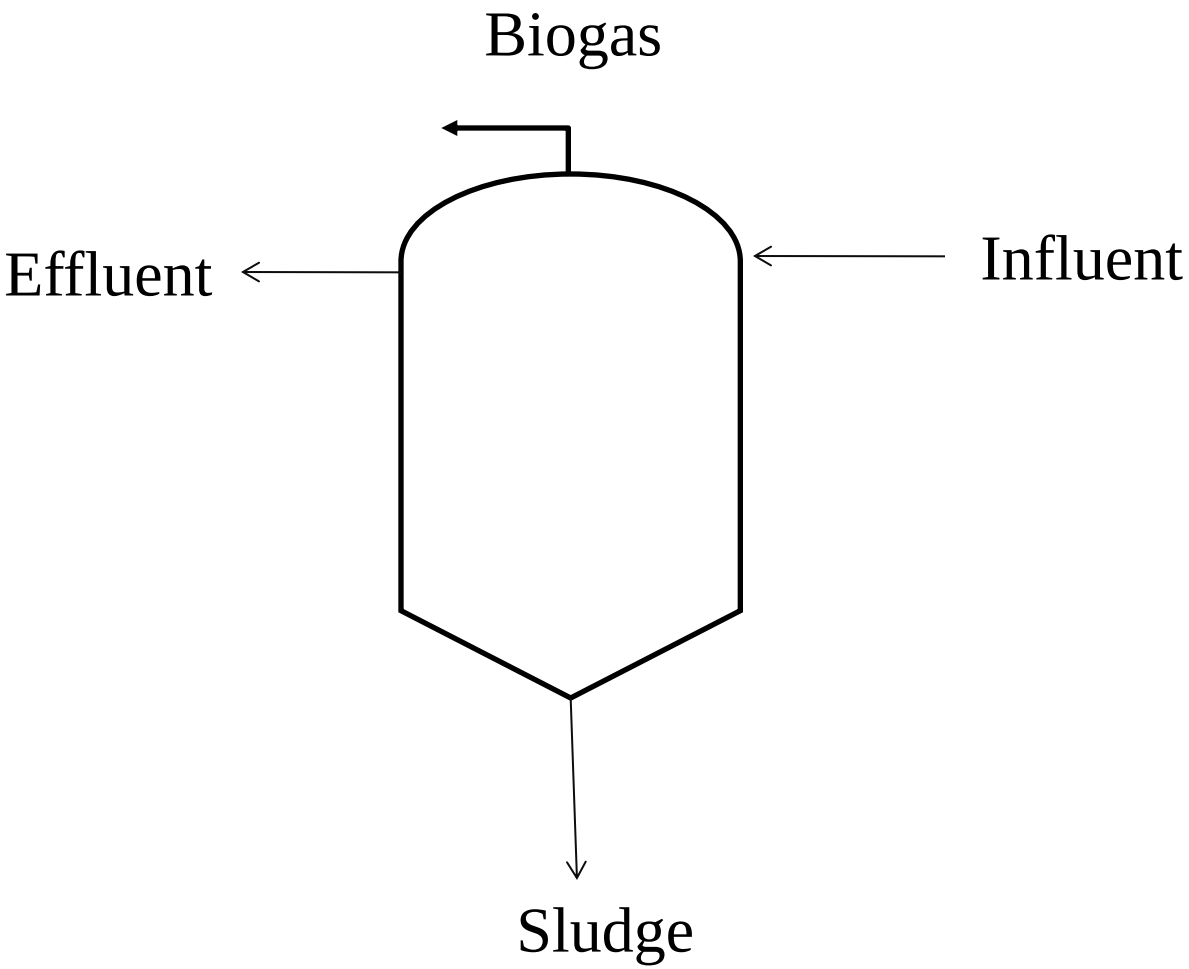
- Anaerobic digestion was the mostly tried option: anaerobic digester, diphasic anaerobic digester, UASB, Fluidized bed anaerobic filter, etc.



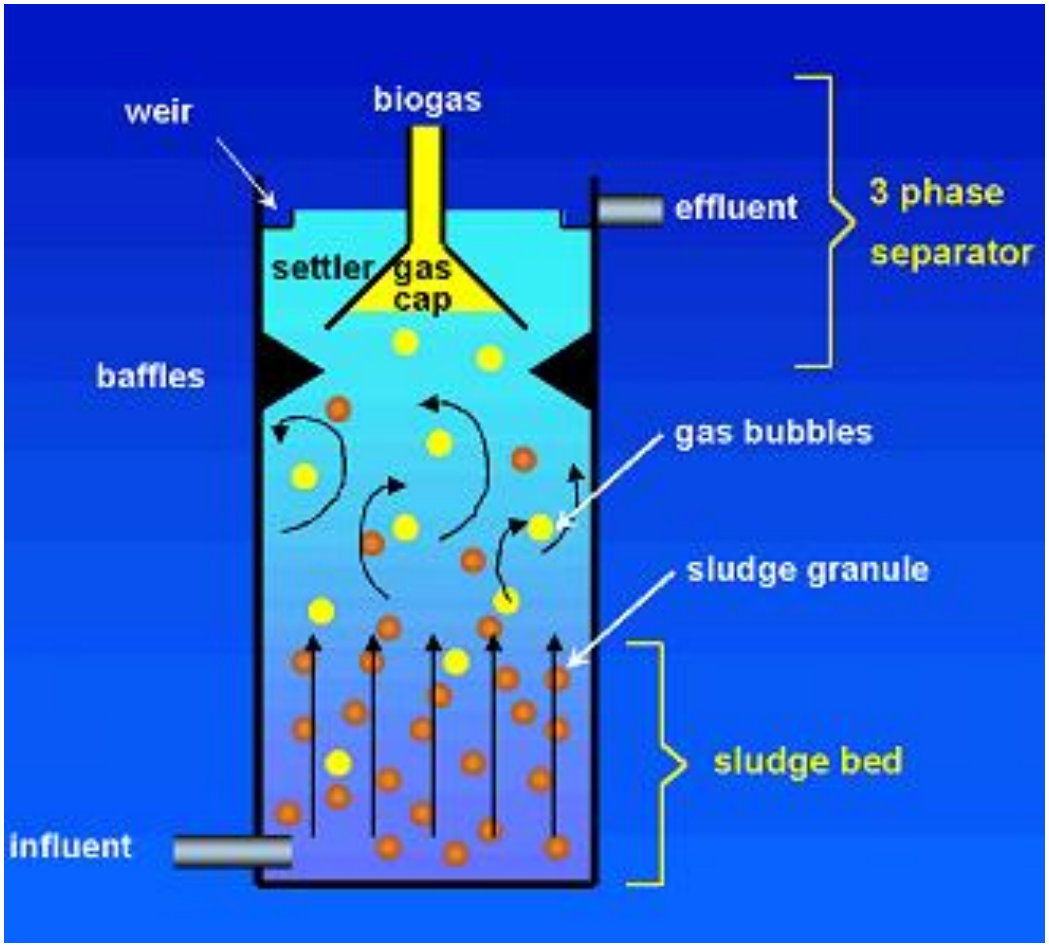
This effluent cannot be disposed off to a river or sewer line or ocean

Disposal Standard = 30 mg/L for disposal into surface waters

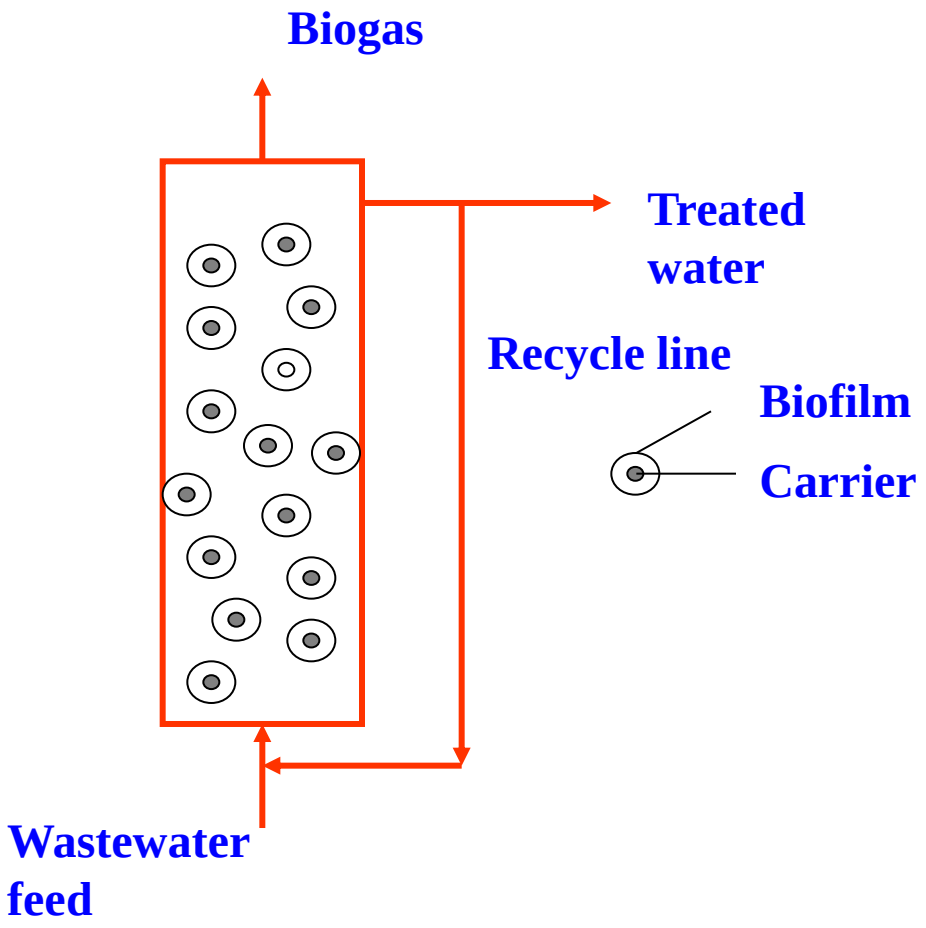
Conventional Anaerobic Digester



UASB (Upflow Anaerobic Sludge Blanket) Reactor

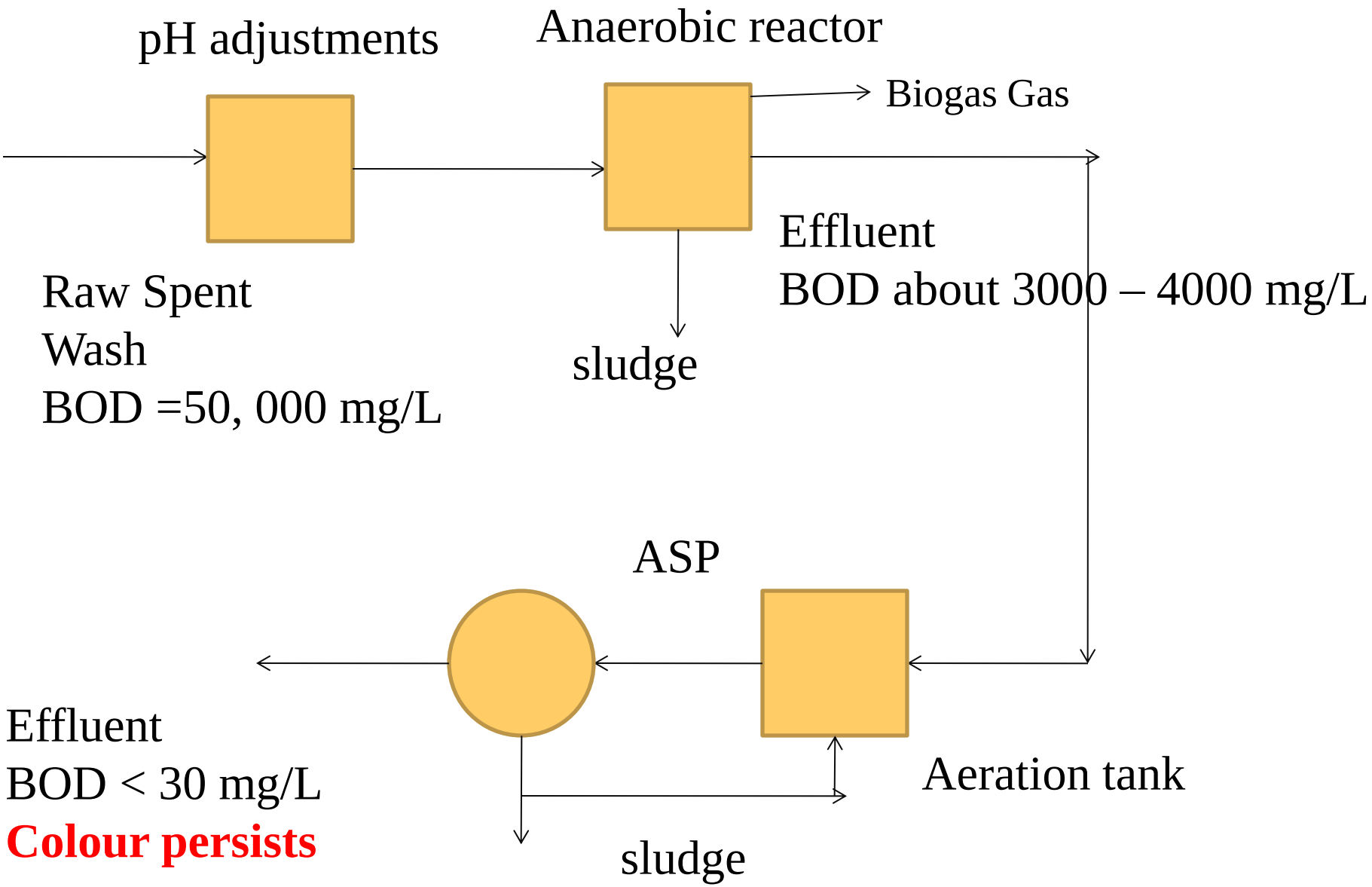


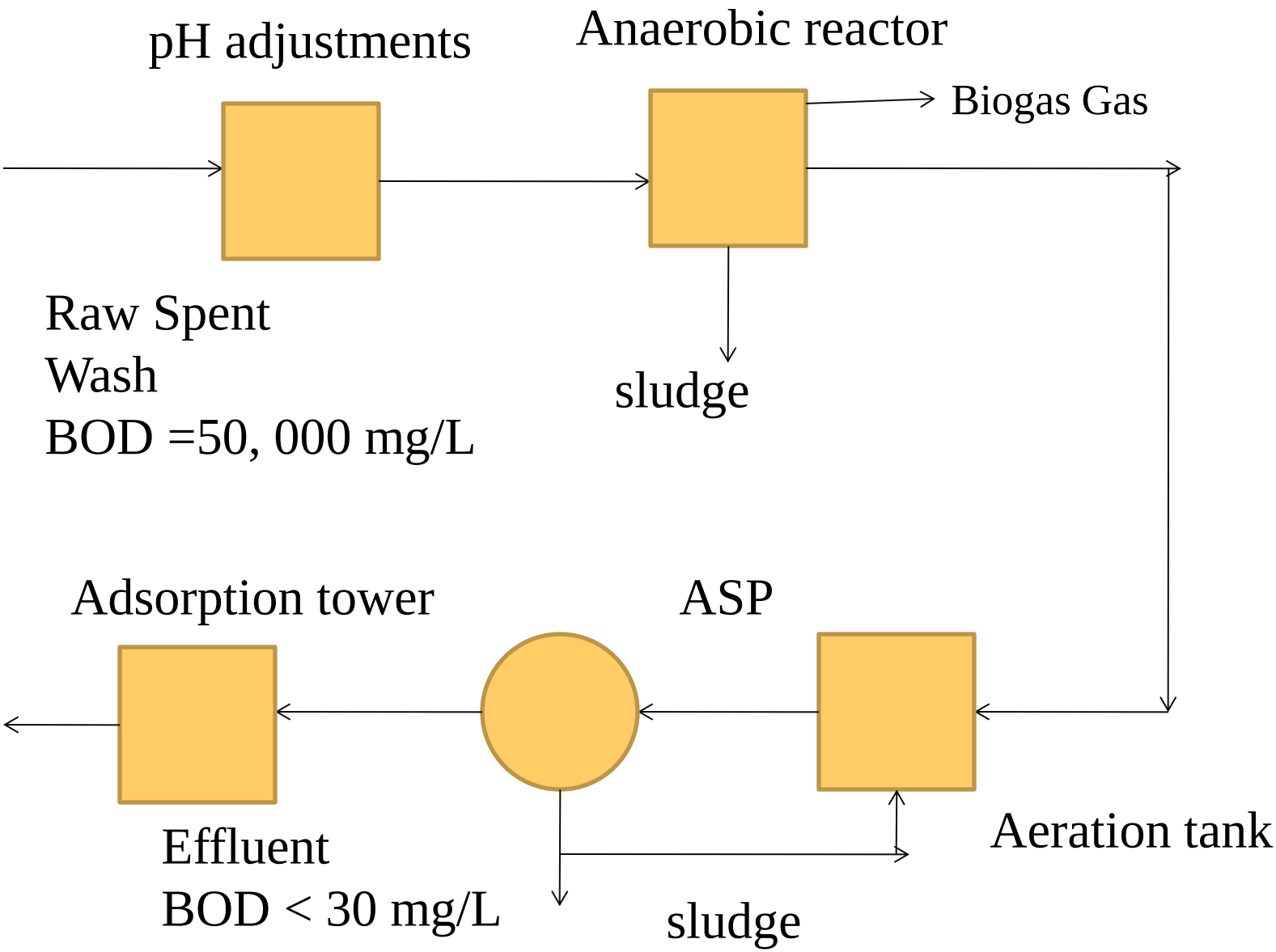
Anaerobic Fluidized Bed Reactor (AFBR)



- Anaerobic digestion bring the BOD down to about 2000 – 4000 mg/lit
- But the discharge standards are normally much lower (30 or 20 mg/lit)
- Hence normally aerobic systems are used to bring the BOD down to < 30 or 20 mg/lit

- Hence the process becomes costly


pH adjustments
Anaerobic reactor
Biogas Gas
Raw Spent
Wash
BOD =50, 000 mg/L
sludge
Effluent
BOD about 3000 – 4000 mg/L
ASP
Effluent
BOD < 30 mg/L
Colour persists
sludge
Aeration tank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Wash
BOD =50, 000 mg/L
sludge
Adsorption tower
ASP
Effluent
BOD < 30 mg/L
sludge
Aeration tank

Sludge Treatment

- Sludge is usually concentrated using a filter press or in a vacuum unit and the sludge cake produced

or

- Sludge can be dried in sludge drying beds (solar energy) and sludge cakes produced

- Sludge cake can be used as manure for horticulture, if it is safe. Or else, to be disposed off in landfills.

Problems with Biological Treatment of Spent Wash

- Requirement of ‘polishing’
- Uncertainity involved with biological systems
- Influence of external parameters like weather, temperature
- Requirement of energy intensive secondary treatment

- Colour problem still persists
- CH_4 generated in the first step is used in the subsequent steps
- Much slower than thermal systems
- More space/volume required

Thermal Treatment of Spent wash

- There have been many other attempts to deal with spent wash
- Burning of concentrated spent wash dates back to world war 1 period (not in India)
- Production of activated carbon, Solid CO2, potash, etc was one of them (reported by Reich in 1945 from US)

Thermal properties of Spent wash

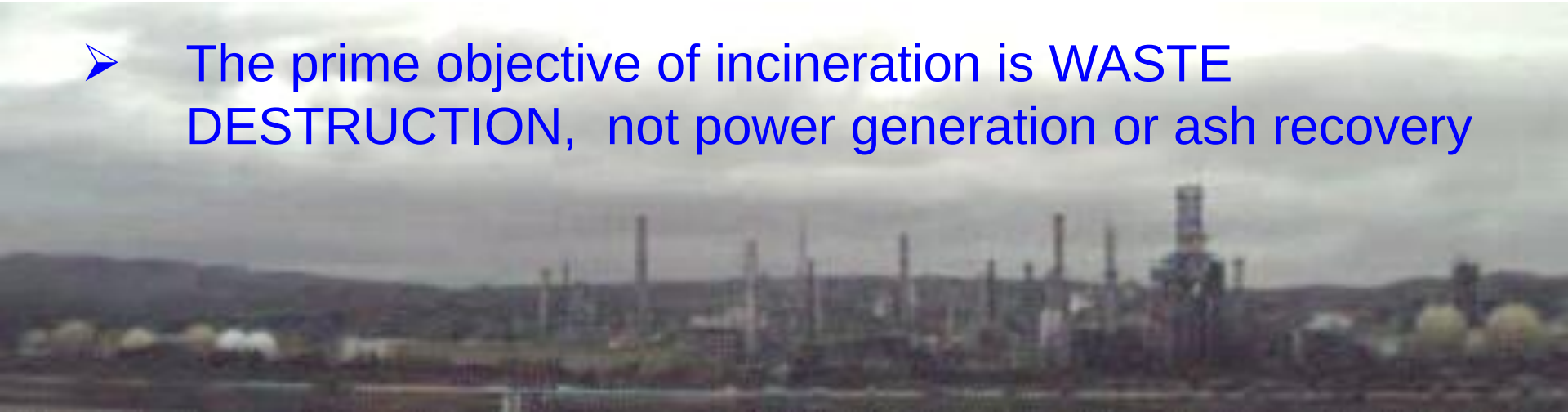
- Sp gr 1.04 to 1.05
- Total Solids: 10 to 22 %
- Total VS: 6 to 8 %
- Ash content: 2 to 3 %
- HHV: 3200 to 3800 kCal/kg (dry solids)
- Density of concentrated spent wash 1.350 g/cc

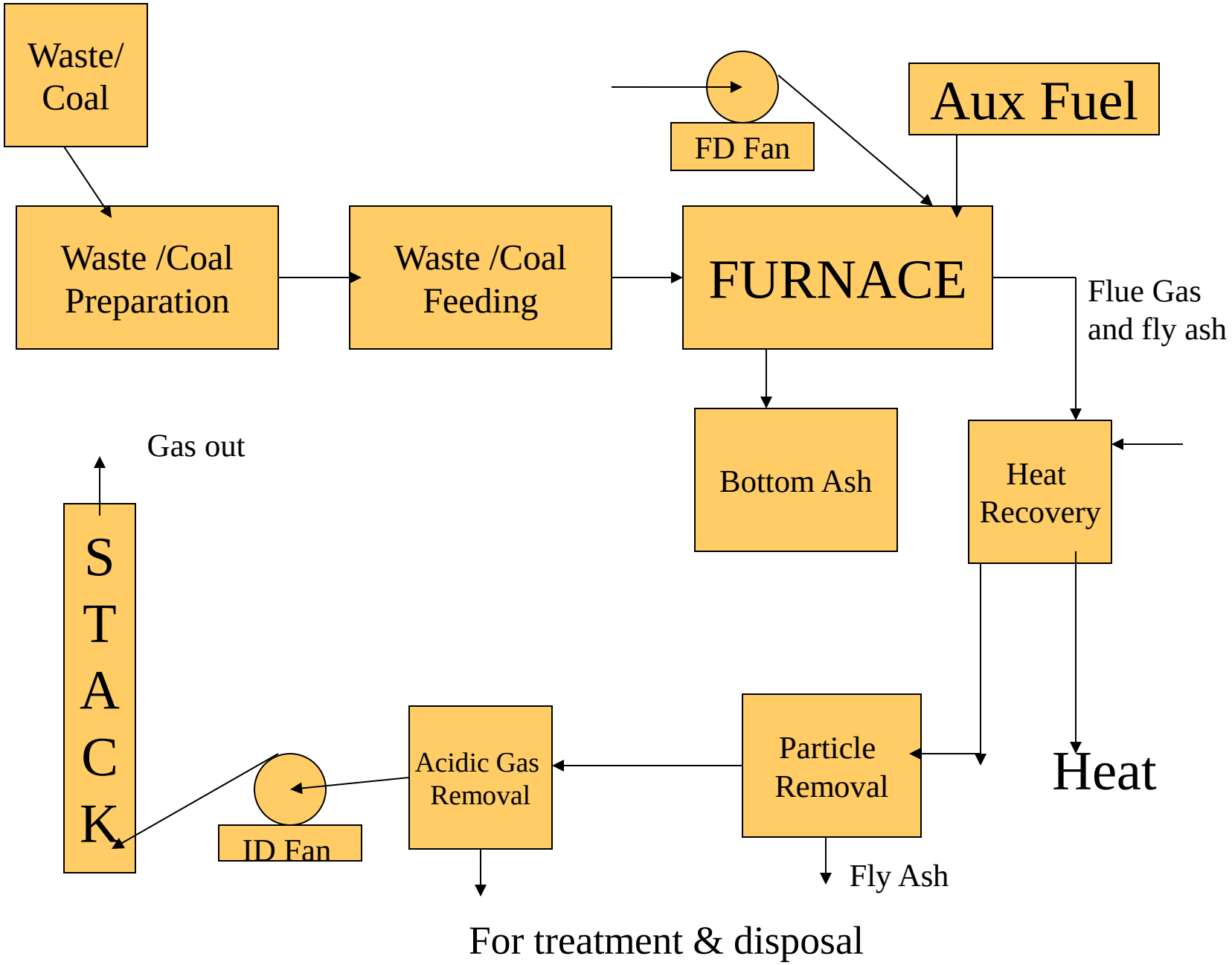
PROCESS OF INCINERATION

- Combustion (controlled burning) of wastes in properly designed and constructed furnace to sterile ash with proper care for air pollution and water pollution.

- It is proper to say INCINERATION SYSTEM rather than simply ‘incinerator’

- The prime objective of incineration is WASTE DESTRUCTION, not power generation or ash recovery



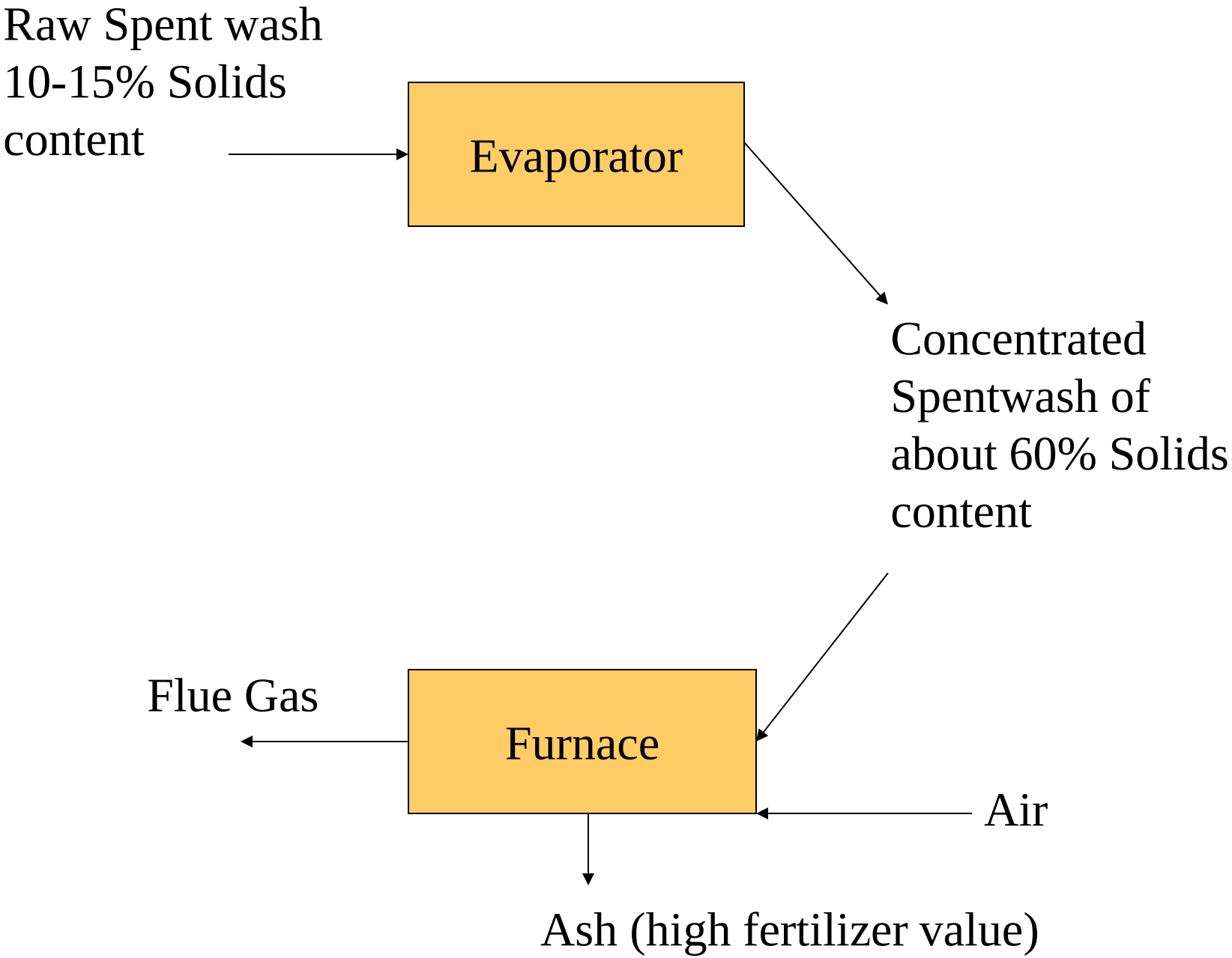

Waste/
Coal
FD Fan
Aux Fuel
Waste /Coal
Preparation
Waste /Coal
Feeding
FURNACE
Flue Gas
and fly ash
Bottom Ash
Heat
Recovery
Gas out
S
T
A
C
K
Acidic Gas
Removal
Particle
Removal
Heat
ID Fan
Fly Ash
For treatment & disposal

Air Pollution Control Equipment

Equipment	Pollutant Removed	% Removal	Pressure Drop Inches of Water Column
Settling Chamber	Large particles	0 to 30 %	0.5 to 1
Multiple Cyclones	Particles	30 to 80 %	3 to 4 (single cyclone: about 1)
Venturi Scrubber	Particles & acidic gases like HCl, SO2	80 to 96 %	10 to 50
Packed scrubber	Aerosols & acidic gases	80 to 96 %	6 to 8
ESP	Fine particles	90 to 97 %	0.5 to 1
Fabric Filter bag House	Fine particles	97 to 99.9 %	1 to 8

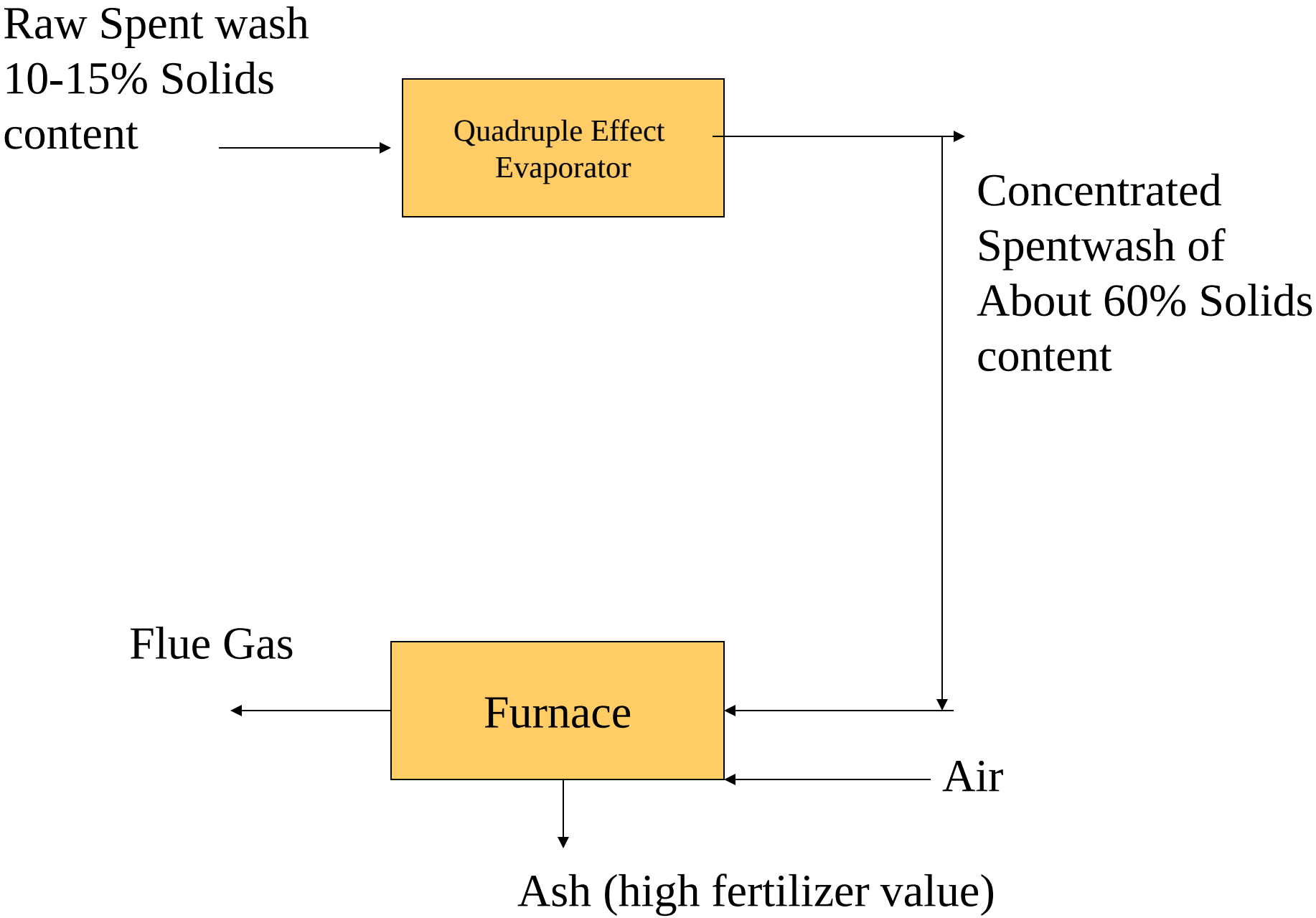
Incineration of Distillery Spentwash: Indian Experience

- First incineration was reported in early nineteen sixties, but not popular
- Became popular in late nineteen eighties. Two popular designs were
  - Destrotherm from Thermax, Pune
  - Sprannihilator from Praj Consultech, Pune


Raw Spent wash
10-15% Solids
content
Evaporator
Concentrated
Spentwash of
about 60% Solids
content
Flue Gas
Furnace
Air
Ash (high fertilizer value)

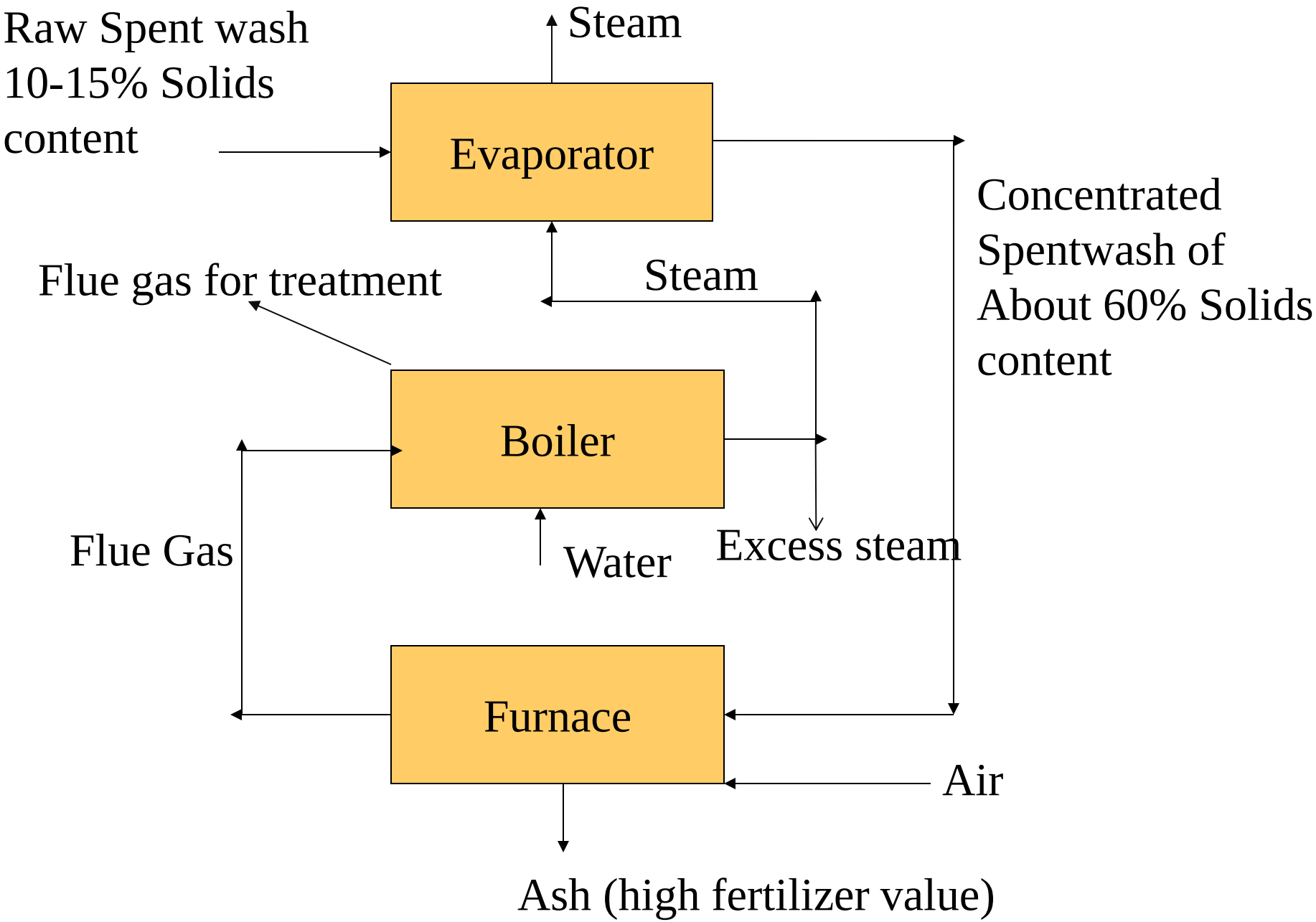
1st incineration plant for spentwash

- Set up at Dyer Meakin Breweries Ltd., Lucknow in 1960
- Capacity: 90 gal spent wash / hour
- Uses forced circulation evaporator for concentrating spentwash to 60% solids content
- Ash from spentwash incinerator: contains high Potash as K_2O (37% of the ash)


Raw Spent wash
10-15% Solids
content
Quadruple Effect
Evaporator
Concentrated
Spentwash of
About 60% Solids
content
Flue Gas
Furnace
Air
Ash (high fertilizer value)

Destrotherm from Thermax

- Govt. Distillery at Chitali, Maharshtra in 1984
- Expected performance for a 45KLD distillery
  - Spent wash generation: 3000lit/hr
  - Ash collection: 700 kg/hr
  - Power generation: 700kW (Process electricity consumption 500 kW)
  - Steam generation: 11515 kg/hr (40bars, 400oC) and steam consumption 5500kg.hr (at 5 bars)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10-15% Solids
content
Steam
Evaporator
Concentrated
Spentwash of
About 60% Solids
content
Flue gas for treatment
Steam
Boiler
Flue Gas
Water
Excess steam
Furnace
Air
Ash (high fertilizer value)

DESTROTHERM
PILOT PLANT FLOW DIAGRAM

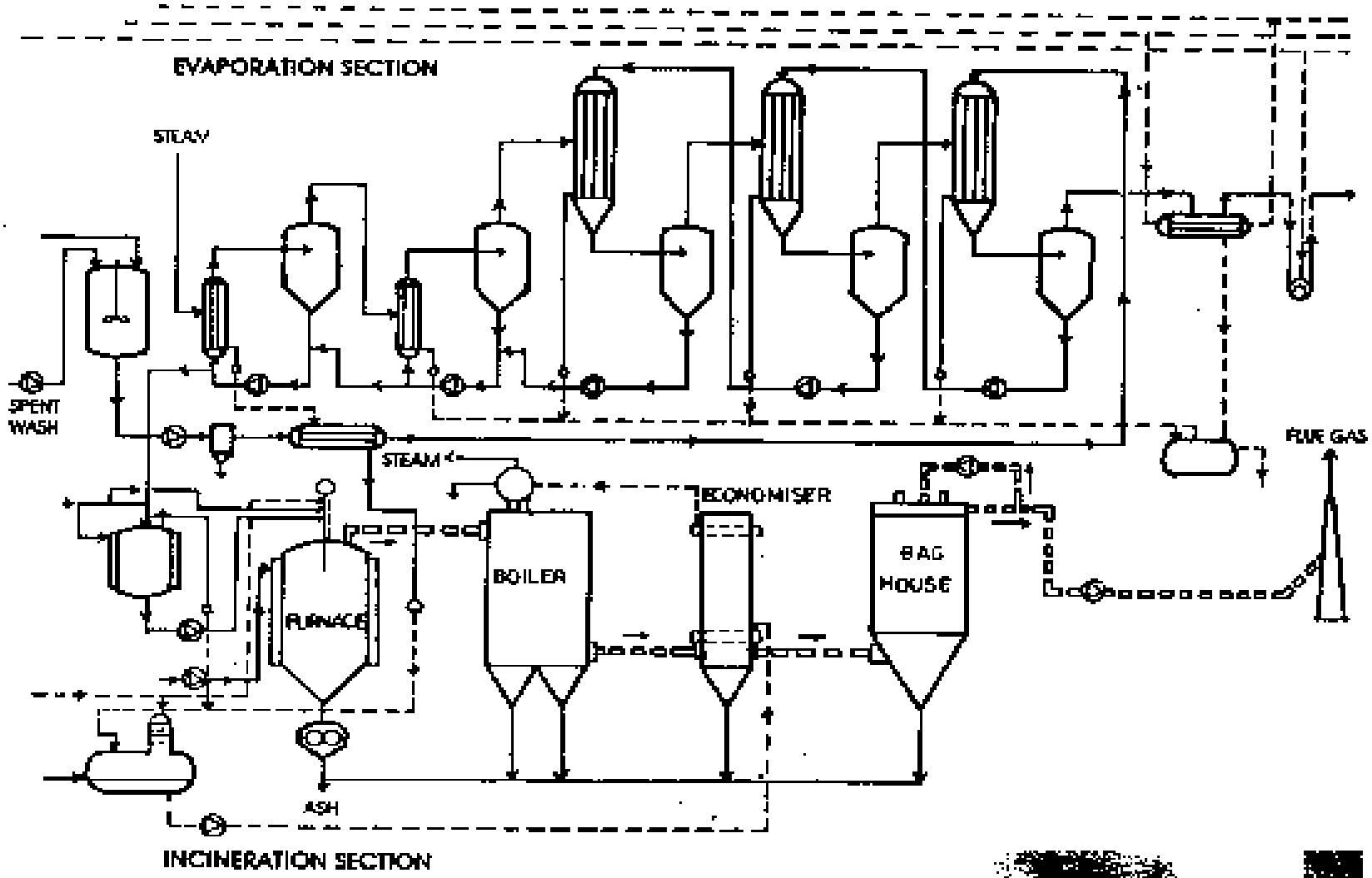


Fig .2·5. Destrotherm (Basargekar, 1989)

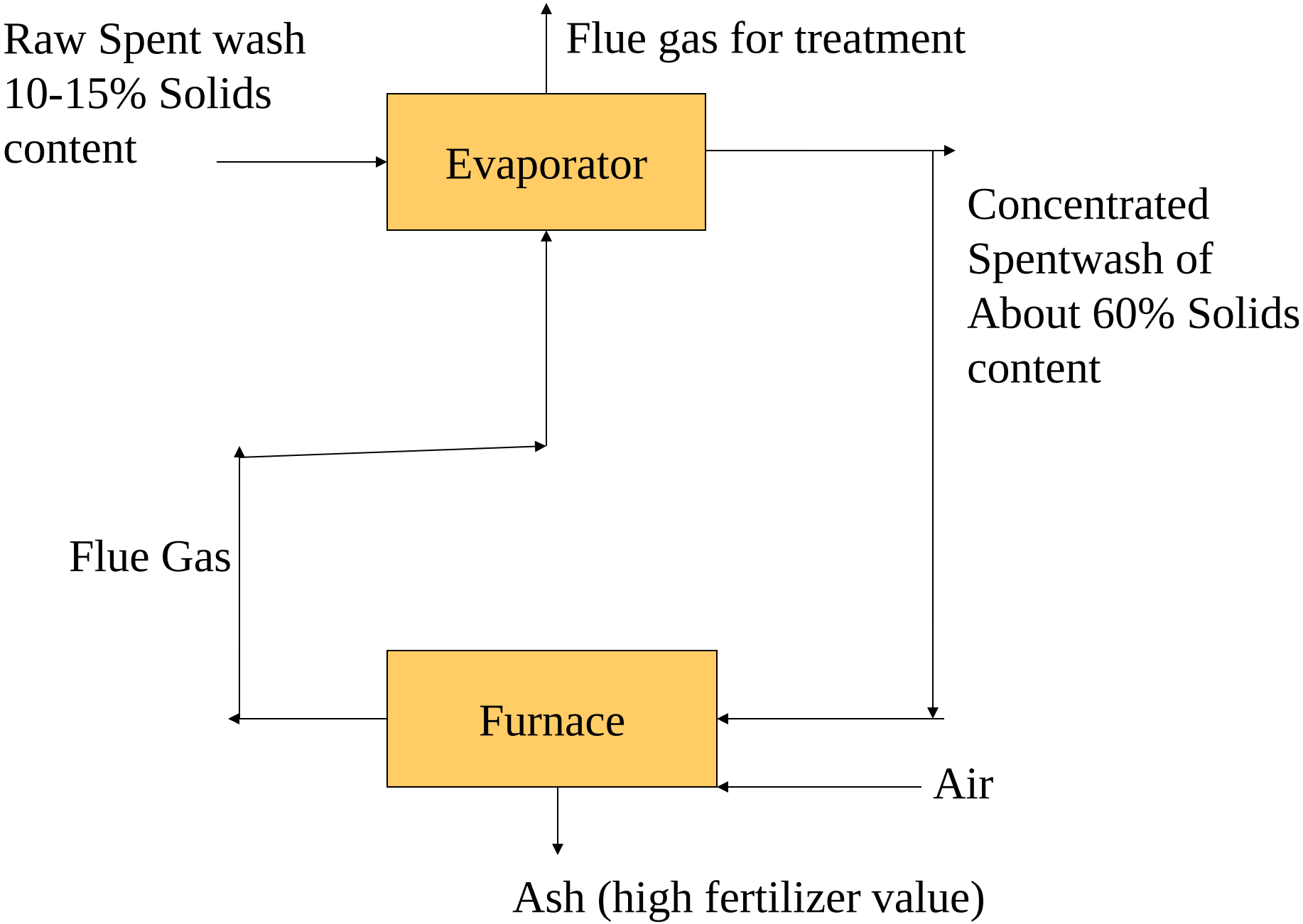
Destrotherm

- Spent wash of 60% solids content is sprayed into the furnace
- Raw spent wash is concentrated to 60% in external forced draft evaporators using process steam
- The flue gas generated goes for the production of steam
- Ash is rich in potash

- Destrotherm did not pick up the expected popularity may be due to
  - The complexity involved in the evaporation system
  - Large area / volume occupied by the evaporators
  - Clinker formation on the boiler tubes
  - High cost involved

Sprannihilator from Praj Counseltech

- Aimed at solving some of the problems with Destrotherm
- Evaporation section made simple and easy to handle
- Plants becomes more compact
- First plant at Kohlapur (M/S. Gadhinglaj Taluka Sahakari Sakhar Karkhana Ltd.)


Raw Spent wash
10-15% Solids
content
Flue gas for treatment
Evaporator
Concentrated
Spentwash of
About 60% Solids
content
Flue Gas
Furnace
Air
Ash (high fertilizer value)

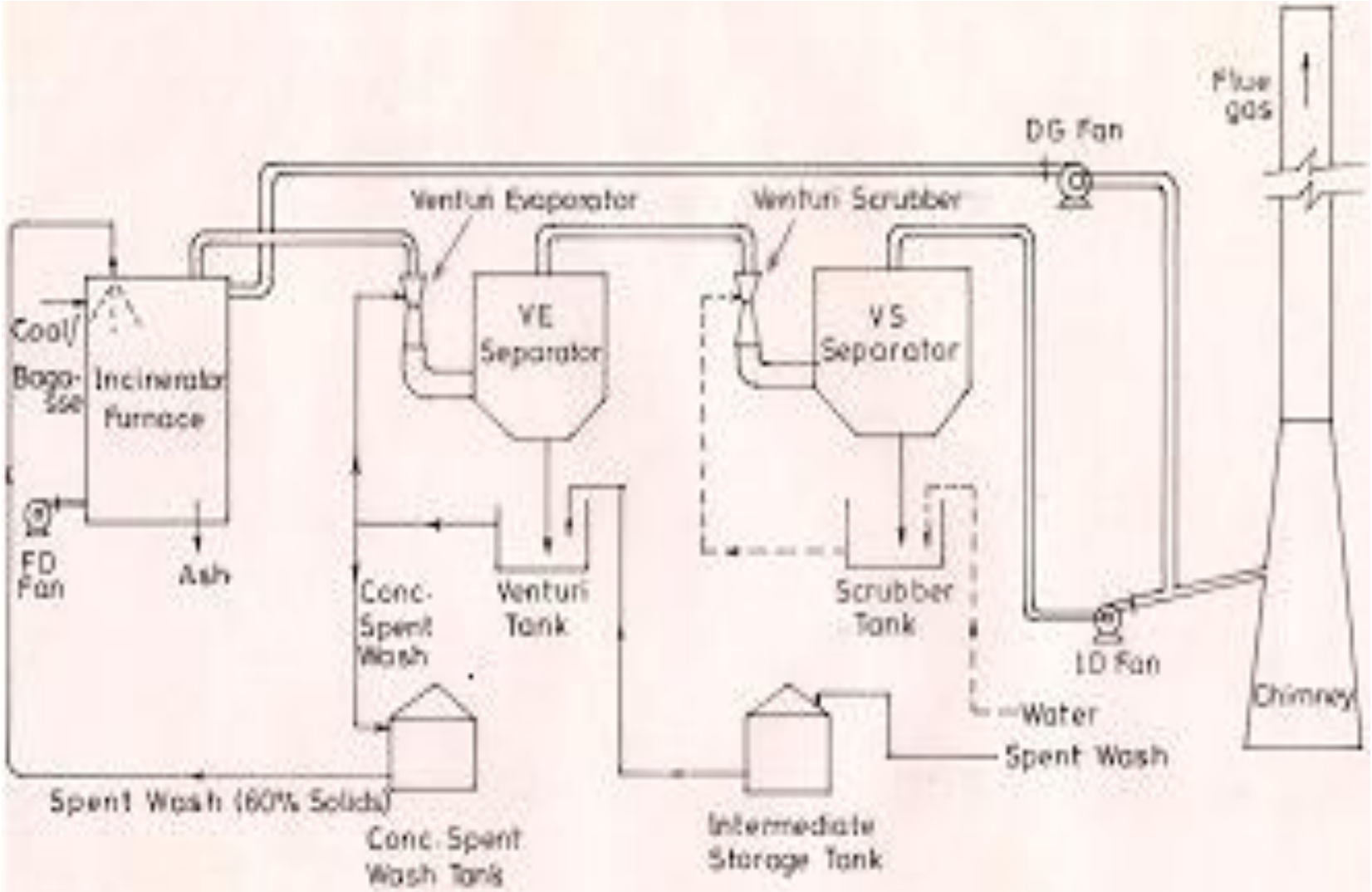


Fig. 2.4. Sprannihilator (Inamdar, 1989)

Sprannihilator

- About 60% solids content spent wash is sprayed into the furnace

- Raw spent wash is concentrated to 60% using the flue gas heat in a venturi-evaporator
- Ash is rich in potash

- More units came up soon at a Cost of about Rs. 37,000,000 (37 Million)

- One at M/S. Polychem Ltd., Nira
  - 45KLD distillery

- Another at Liquors India Pvt. Ltd., Hyderabad
  - 20 KLD distiller

Problems with Incineration of spent wash

- Solids content of spent wash is to be brought to about 60% before firing into the furnace.
- This brings the economics down as it is energy intensive. Also, damaging to the material of construction.
    - Spent wash is sticky
    - It swells (up to about 5 times the original volume) while heating

Protocols/ Guidelines related to Treatment & Disposal options for distillery effluent

- **Existing Molasses based Distilleries: Zero Effluent Discharge (100% by Dec. 2005) through**
  - **Bio-composting,**
  - **Concentration and Incineration**
  - **Biomethanation followed by two stage secondary treatment followed by ferti-irrigation**
  - **Biomethanation followed by pre-sown irrigation**
  - **Biomethanation followed by treatment and Discharge into deep sea.**

- **New Distilleries & Expansion of Existing Distilleries (Molasses based)**
  - **Zero Effluent Discharge in surface water/ ground water**

Protocols/ Guidelines related to Treatment & Disposal options for distillery effluent

CPCB Resolution (June 2008)

- **New stand alone Distilleries:**
  - **Concentration – incineration system**
- **New attached Distilleries:**
  - **Bio-methanation followed by bio-composting**
  - **Concentration– incineration system {Reboiler/ Evaporation/ Concentration followed by incineration of concentrated spent wash in boiler (for power generation)}**
- **Existing stand alone Distilleries (expansion):**
  - **Concentration – incineration system**

Protocols/ Guidelines related to Treatment & Disposal options for distillery effluent

CPCB Resolution (June 2008)

- Distilleries (both stand alone and attached) not complying with the required environmental standards to switch over to emerging technologies from existing technologies of Ferti-irrigation, One time land application and composting in a time bound manner.

Co-Processing of Distillery Spent wash concentrate in Cement Industry (CPCB, 2010)

Co-processing of molasses based distillery spent wash concentrate in Cement Kilns, furnaces of Thermal Power Plants/ Steel Plants, Incinerators

Concentrating spent wash

- About hundred RO (Reverse Osmosis) plants working in distilleries

- About hundred MEE (Multiple Effect Evaporation) system working in distilleries

Reverse Osmosis

Raw Spent wash
10-15% Solids
content

Concentrated
Spentwash of
50-60% Solids
Content to
Boilers / Incinerators
Cement Kilns / Drying

MEE (Multiple Effect
Evaporator)

Co-Processing

- Co-processing in cement industry refers to the substitution of primary fuel and raw material by waste, recovering energy and material from waste.

- Waste materials used for co-processing are referred to as alternative fuels and raw materials (AFR).

- Raw material for Cement manufacturing: Lime stone & Clay / shale: Calcium Carbonate, Alumina, Silica, Oxides of iron)

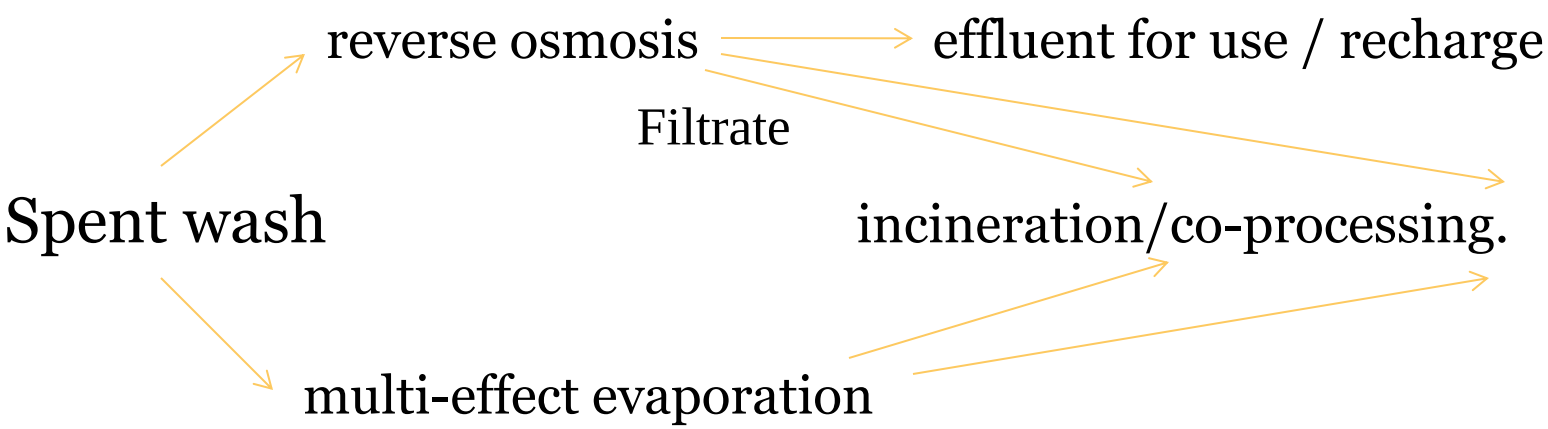
Cement Kiln Suitability

- High temperatures (1400 ^{O}C) and residence time of 4 – 5 seconds in an oxygen–rich atmosphere ensure the destruction of organic compounds.
- Any acid gases formed during combustion are neutralized by the alkaline raw material and are incorporated into the cement clinker.
- Interaction of the flue gases and the raw material present in the kiln ensures that the non–combustible part of the residue is held back in the process and is incorporated into the clinker in a practically irreversible manner.
- No waste is generated that requires subsequent processing.

Benefits of Co-processing

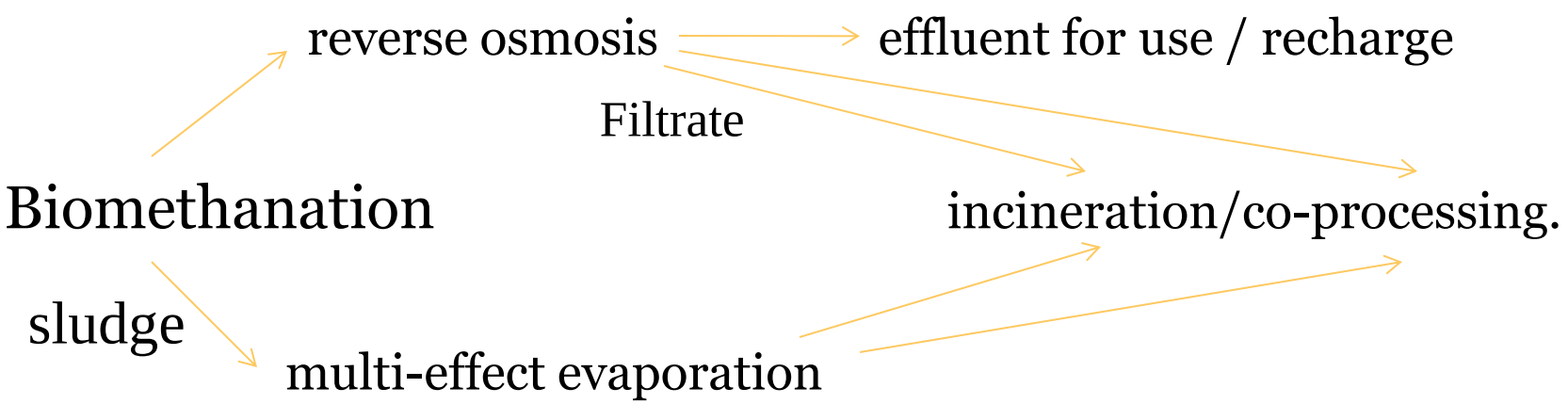
- **Reduction in Green House gases emission & related benefit of carbon trading**
- **Conversion of waste into energy / as a raw mix component**
- **Conservation of fossil fuel resource**
- **Reduction in energy / cement production costs**

Different treatment technology routes for ZERO LIQUID DISCHARGE (ZLD)



- Concentration through RO/ MEE followed by incineration/ co-processing.

Different treatment technology routes for ZERO LIQUID DISCHARGE (ZLD)



- Concentration through RO/ MEE followed by incineration/ co-processing.